

Development and Preliminary Validation of the Enjoyment Questionnaire and the Sources of Enjoyment Questionnaire

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ABSTRACT

Building on the sources of enjoyment identified by Schaffer and Fang's card sorting study, we describe the development of two new measures for player experience research, the Enjoyment Questionnaire (EQ) and the Sources of Enjoyment Questionnaire (SoEQ). Among other sources, the EQ and SoEQ draw on flow theory, self-determination theory, and desire fulfillment theory. The EQ assesses digital game enjoyment and the SoEQ assesses 38 sources of enjoyment in digital games, including Humor, Relaxation, Savoring, Optimal Pacing, Optimal Variety, Social Responsibility, Task Significance, and Clear Task Purpose. The scale was validated with a survey of 564 participants. Results demonstrated the EQ and each subscale of the SoEQ have both construct validity and internal consistency. Results also provided evidence for convergent and discriminant validity. The EQ and SoEQ are useful and reliable tools for the study of digital game enjoyment and its sources.

CCS CONCEPTS

• Human-centered computing; • Empirical studies in HCI; • HCI theory, concepts and models; • Applied computing; • Computer games;

KEYWORDS

Enjoyment, sources of enjoyment, factor analysis, scale development, flow, intrinsic motivation, desire fulfillment, game design, gamification, serious games, digital games

ACM Reference Format:

Owen Schaffer. 2022. Development and Preliminary Validation of the Enjoyment Questionnaire and the Sources of Enjoyment Questionnaire. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '22 Extended Abstracts)*, April 29–May 05, 2022, New Orleans, LA, USA. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3491101.3519819>

1 INTRODUCTION

We define enjoyment here as the positive evaluation of an experience. The sources of enjoyment are therefore defined here as positive experiences that may contribute to that positive evaluation we call enjoyment. To reliably make games and other interactive

designs more enjoyable, Human-Computer Interaction (HCI) researchers need more comprehensive measures of the sources of enjoyment. Schaffer and Fang [46] reduced 167 sources of enjoyment from the literature to 34 categories they called the sources of enjoyment in digital games with an iterative card sorting method with game players. However, these sources of enjoyment had yet to be developed into a psychometrically-valid and freely-available questionnaire that researchers and practitioners can use to improve user experiences. The present research fills that gap in the literature and presents Enjoyment Questionnaire (EQ) and the Sources of Enjoyment Questionnaire (SoEQ) as new measures of player experience, and a shortened version of the same measures.

Developing more comprehensive measures of enjoyment and the sources of enjoyment is important because many of the popular existing measures of player experience do not include potential sources of enjoyment identified by Schaffer and Fang [46], such as Humor, Relaxation, and Savoring to name a few. This new measure allows these factors to be explored in future research and leveraged by practitioners to design for enjoyment. Since many of the sources of enjoyment Schaffer and Fang [46] identified were drawn from the literature, the EQ and SoEQ build on and incorporate previous theories of positive experience. For example, these measures include factors drawn from Flow Theory [31, 45], Self-Determination Theory [7, 40], and Reiss and Havercamp's Theory of Basic Human Desires [18, 39, 44].

This paper also presents the testing and preliminary validation of the EQ and SoEQ. Questionnaire items were developed with a combination of items drawn from existing measures where possible and original items devised following DeVellis and Thorpe's guidelines for scale development [2:Chapter 5]. The items are then tested, first with a pilot study of 293 participants and then with a survey of 564 participants, to test the reliability and validity of the measures. Participants were asked assess the last time they played a digital game using the initial proposed items. The resulting measure is presented and discussed.

2 RELATED WORK

Sources of enjoyment are positive experiences that may contribute to enjoyment. Enjoyment is defined here as the extent to which people positively evaluate their experience. Three theories that played an important role in the development of the Sources of Enjoyment Questionnaire are Flow Theory, Self-Determination Theory, and Reiss and Havercamp's Theory of Basic Human Desires.

Flow is the experience of overcoming challenges for the sake of the enjoyment they provide. Flow theory says that three flow conditions lead to the flow state – optimal challenge, clear proximal goals, and immediate progress feedback – and that flow state in turn leads to enjoyment [31, 45]. Nakamura and Csikszentmihalyi described

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CHI '22 Extended Abstracts, April 29–May 05, 2022, New Orleans, LA, USA

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ACM ISBN 978-1-4503-9156-6/22/04...\$15.00

<https://doi.org/10.1145/3491101.3519819>

the flow experience as “engaging just-manageable challenges by tackling a series of goals, continuously processing feedback about progress, and adjusting action based on this feedback” [31:240]. Jackson and Marsh [20] developed the Flow State Scale to measure the flow factors in sports, and Payne et al. [32] adapted it to create a more general Activity Flow State Scale.

Self-Determination Theory (SDT) says that fulfilling psychological needs for competence, relatedness, and autonomy leads to intrinsic motivation and enjoyment [7, 40]. Ryan, Rigby, and Przybylski (2006) created the Player Experience of Need Satisfaction (PENS) model. PENS began with SDT, but added Presence, feeling like you are there in the game or are one of the characters, and Intuitive Controls, how much the game’s controls make sense and are easily mastered.

Reiss and Havercamps’s theory of basic human desires proposed that there are 16 basic human desires that drive human motivation [18, 38, 39]. Examples include Idealism, Romance, Power, and Tranquility. Havercamp [18] developed a measure of those 16 basic human desires, and Schaffer [44] adapted that measure to the study of digital game enjoyment.

Yee [50] proposed and tested a model that three factors motivate people to play online games: achievement, social, and immersion. Yee assessed the construct validity of this model with factor analysis on data from a large-sample survey. Quick, et al. [36] created a six-factor taxonomy of game enjoyment and validated it with factor analysis of survey data: Fantasy, Exploration, Fidelity, Companionship, Challenge, and Competition.

Brown and Vaughan [3] proposed eight play personalities: The Joker, The Kinesthete, The Explorer, The Competitor, The Director, The Collector, The Artist/Creator, and The Storyteller. Fullerton [14] expanded on those play personalities, calling them player types and adding The Performer and The Achiever.

Lazzaro [23, 24] proposed four pathways to emotion in games: Easy Fun (Novelty & Curiosity), Hard Fun (Challenge & Fiero), People Fun (Friendship & Amusement), and Serious Fun (Altered States & Relaxation). Lazzaro [23] also identified some of the emotions people experience when they play games, such as Fear, Surprise, *Naches* (Yiddish for enjoying the accomplishments of mentees), *Fiero* (Italian for triumph and pride), and *Schadenfreude* (German for enjoying the pain of others). Korhonen, Montola, and Arrasvuori, [22] created the playful experiences or PLEX framework, which was made up of 20 categories of playful experiences, including Completion, Discovery, Relaxation, Sensation, Expression, Subversion, and Fellowship. Lucero and Arrasvuori [26] developed a set of PLEX cards with one of those playful experiences on each card.

The present study builds on the sources of enjoyment identified by Schaffer and Fang [46], who conducted an iterative card sorting study with sixty gamers, or participants who reported playing games at least once per week. The starting place for Schaffer and Fang’s card sorting study was 167 sources of enjoyment drawn from the literature which included the sources of enjoyment discussed above in this section. The participants sorted cards into categories to reduce 167 sources of enjoyment drawn from the literature to 34 final categories of the sources of enjoyment. After every 10 participants, inter-rater agreement between participants was evaluated with Randolph’s multi-rater kappa [37] and the cards and categories were revised. The first 40 participants were also able to put

cards in more than one category, create new categories, and sort cards into a “Not a Potential Source of Computer Game Enjoyment” category. This card sorting procedure was similar to the method for measure development used by Moore and Benbasat [30], which has been used to create new measures of enjoyment and flow in computer games [11, 12]. The sources of enjoyment that resulted from this iterative card sorting with gamers were grounded in participants’ experience playing games. The present research builds on those sources of enjoyment by creating psychometric measures to assess enjoyment and the sources of enjoyment, and conducting preliminary validation of those measures.

3 METHODS

3.1 Item Development

The sources of enjoyment from Schaffer and Fang’s [46] card sorting study were used as a starting point for item generation. Following DeVellis and Thorpe’s guidelines for scale development [9:5], we devised 6-22 items for each construct or source of enjoyment. This follows their recommendation to starting with an item pool at least 50% larger than the intended size of the final number of items per construct because our target was 4 items per construct. This process was guided by the definitions Schaffer and Fang [46] gave for the sources of enjoyment they identified. Where existing previously validated measures could be found in the literature that fit well with those sources and their definitions, those items were included, and where items could not be easily found original items were generated by following DeVellis and Thorpe’s procedures for item development. All items were statements participants responded to with 7-point Likert scales of agreement ranging from “Strongly Disagree” to “Strongly Agree” with all scale points labeled. The full list of initial items and their sources can be found in Supplement A.

Schaffer and Fang [45] distinguished between flow conditions, the factors that lead to flow, and flow indicators, the factors that indicate how much a person is experiencing flow. Only the flow conditions Clear Proximal Goals, Immediate Progress Feedback, and Optimal Challenge from flow theory were included in the sources of enjoyment from Schaffer and Fang’s [46] card sorting study. But the flow indicators may be sources of enjoyment as well. For example, Sackett et al. [42] found that feeling time was flying by faster lead to enjoyment rather than the other way around. So, the flow indicators and items for them were included as potential sources of enjoyment: Altered Perception of Time, Loss of Self-Consciousness, Merging of Action and Awareness, Sense of Control. Clear Navigation, defined here as the experience of knowing where to go next throughout an activity, extends the flow condition of Clear Proximal Goals, which is knowing what to do next through each step of an activity, by applying it to the task of navigating the space of a game world.

While generating items for one of the sources of enjoyment from Schaffer and Fang’s [46] card sort, “Significance, Meaning, Purpose, and Legacy”, two constructs emerged and were included based on theory: Task Significance, which we define as feeling that what you are doing is important or meaningful, and Clarity of Task Significance, which we define as knowing why your actions are important or meaningful. These are conceptually similar to the concepts of Significance and Purpose from Mekler and Hornbæk’s [29] framework for the experience of meaning in HCI, but to the

best of our knowledge they have yet to publish a measure of these concepts. Another construct similar to Task Significance, Stakes and Investment is defined here as feeling that the stakes are high or the sense that one has a lot to lose or gain from succeeding or failing at a task. Stakes and Investment was included as a potential source of enjoyment because feeling that the stakes are high in a game could increase one's enjoyment of that game.

Items for several sources of enjoyment were adapted from Schaffer's [44] measures of desire fulfillment for digital games based on Reiss and Havercamps's theory of basic human desires, which proposed that 16 basic human desires comprise the underlying motivations for human behavior [18, 38, 39]. Items for fulfilling a desire for Idealism were used as potential items to measure Task Significance, items for fulfilling a desire for Honor were used as potential items to measure Self-Worth and Integrity, and items for fulfilling a desire for Organization were used as potential items to measure Strategizing, Problem Solving, and Critical Thinking. In the factor analysis later, these items became their own constructs: Social Responsibility, Honor, and Organizing. Each of these was included because they were potential sources of enjoyment that were not well explained by the other sources of enjoyment. Enjoyment items were included in the factor analysis with the sources of enjoyment items to develop a measure of Enjoyment for future research and to test and ensure discriminant validity between enjoyment and the sources of enjoyment. In other words, Enjoyment was included in the factor analysis to ensure that none of the sources of enjoyment were themselves measures of enjoyment. Enjoyment items were drawn primarily from the *Intrinsic Motivation Inventory* [28] and the Autotelic Experience flow indicator subscale [20, 32]. The *Player Experience Inventory* [1] was mapped to Schaffer and Fang's [46] sources of enjoyment and all of its items appeared to map onto sources of enjoyment, so all of its items were included as items in the initial measures.

A pilot study with 296 participants was conducted in the same way as described in the main study below with a few key exceptions. First, all of the items in the pilot version of the sources of enjoyment questionnaire were preceded with "While playing this game, I enjoyed. . ." rather than "While playing this game, . . ." After the pilot study we decided to remove the "I enjoyed" part of the item stem. Second, Sense of Control items were excluded from the survey by mistake in the pilot and were included in the main study. Third, the pilot study was conducted with an international sample rather than a US sample, and did not prevent users from using a Virtual Private Network to hide their country of origin. This made it harder to prevent duplicate and fraudulent responses to ensure the quality of the data. This issue was addressed in the main study using the protocol developed by Winter et al. [49].

After the pilot study, some items were reworded for clarity and some items were added to ensure enough items for content validity; there were still 6-22 items per construct. Because Brooke's [2] System Usability Scale (SUS) was used to measure Usability of Controls, two items from that measure appeared to form a separate factor in the pilot study data. This factor was also found by previous factor analyses of the SUS [25] and labeled Learnability. Since this separate factor was supported by prior research, additional items were added to ensure Learnability had enough items to possibly form its own new factor if it survived factor analysis.

We chose to include one additional construct after the pilot, Ability to Retry, based on our experience and observations teaching undergraduate game development courses. Ability to Retry is defined here as having the ability to try again after failing at a task. We observed that student games that provided the ability to retry appeared to be more enjoyable, so we decided to include it so that it could be measured and tested more in future research. The ability to retry failed tasks, such as with frequent opportunities to save one's progress, may have a role in maintaining a flow state by continuing to present players with a series of optimal challenges rather than having to restart the game from the beginning. The full list of initial items and their sources for the measures under development in the present study are included in Supplement A.

3.2 Participants

Five-hundred and sixty-four participants were recruited. Participants were screened to ensure they had played a digital game in the last six months. There were 315 (55.9%) female participants, 244 (43.3%) male participants, and 5 (0.9%) participants who self-identified as non-binary, transgender, gender non-conforming, undecided, or N/A. The participants ranged in age from 20 to 75 with a mean average age of 41.19 years. Participants had played digital games for a mean average of 19.02 years, ranging from 0-55 years. They typically played digital games a mean average of 9.45 hours per week, ranging from 0-56 hours per week. Participants were 18 years or older, US-only, and had an approval rate of at least 97% on at least 1000 tasks on Amazon Mechanical Turk (mTurk).

One benefit of mTurk is that it provides a relatively diverse sample. Buhrmester et al. [5] found that mTurk provided samples that were slightly more demographically diverse than standard internet samples and samples that were much more diverse than typical American college samples. Participants were filtered with CloudResearch's approved participant list to improve data quality. Hauser et al. [17] found that participants on CloudResearch's approved list had better honesty, reliability, reading comprehension, and attentiveness scores than participants on their blocked list or a typical mTurk sample.

3.3 Procedure

Participants were recruited using Amazon Mechanical Turk (mTurk) to fill out an online survey. Participants were presented with the following definition: "A digital game is any game that you play on a computerized device, like a video game console, Personal Computer (PC), smartphone, or on the Internet." Then they were asked to name the last digital game they played. Participants then responded to the questionnaires under development about their experience while playing the game that they named as the last digital game they played. Participants rated each item on a 7-point Likert scale of agreement ranging from "Strongly Agree" to "Strongly Disagree" with all scale points labeled. The last digital game they played was piped or inserted at the top of each page of the survey and before each block of five questions to remind them of the game to which they were responding. Open-ended qualitative questions were asked, for example about what they enjoyed most and least about the game, and participants were asked to type in their answers. The study took approximately one hour, participants

were given \$3.25 USD as an incentive to participate, and the study was approved by the Committee on the Use of Human Subjects in Research at Bradley University, CUHSR # 20-052-E. Hara et al. [16] found Mechanical Turk users received a median of about \$2 per hour, so this is in line with the median pay on this platform.

The Conscientious Responders Scale [27] was used as an attention check, with items such as “Choose the first option – “strongly agree” – in answering this question.” mixed into the questionnaire. These items were randomly mixed into the sources of enjoyment questionnaire questions. Participants were excluded from analysis if they responded incorrectly to attention-checking questions using the Conscientious Responders Scale [27], if they spent less than 20 minutes on the survey, or if their responses to the open-ended qualitative questions were incomplete or did not answer the questions that were asked. The minimum time to complete the survey was determined from the pilot testing, and the median survey completion time in the main study was 54.03 minutes. Participants were also told at the top of each page of the survey that their progress was saved and they could take a break and come back to finish it within one week from when they began to reduce participant fatigue by allowing them to complete the survey over multiple sessions.

4 RESULTS

4.1 Factor Structure and Reliability

Factor analysis was conducted with PROMAX rotation and Maximum Likelihood extraction using IBM SPSS version 27. Factors with an *Eigenvalues* 1.0 or greater were retained. The KMO and Bartlett’s tests showed the data were appropriate for factor analysis (KMO=0.940 or “Marvelous”; Bartlett’s significant at $p<0.001$). Skewness and kurtosis for each observed variable was assessed and did not exceed [3] and [10] respectively [21:76–77]. Items that did not load onto a single factor or that were split across multiple factors were dropped one at a time until a simple factor structure was found. The retained items had factor loadings above [0.4] and any cross-loadings were at least 0.2 less than the main factor loading [15:3,33]. Where there were two items that had very similar wording, the item with the higher factor loading was retained and the other was dropped to reduce unintended redundancy [9:5].

Going through this iterative process of dropping one item at a time and re-running the factor analysis, some items originally intended for separate factors loaded onto the same factor; this is referred to here as those intended factors merging or converging into a new combined factor, which was labeled with a name according to the meaning of its items. Similarly, items intended for a single factor sometimes loaded onto more than one factor, and where that took place this is referred to here as the intended factor splitting into two or more factors, and the new factors were then labeled with new names based on what was common among the meaning of their items. Some items intended for one factor did not load onto a single factor, so removing those items is referred to here as dropping that intended factor. What follows is a description of how the factors merged, split, or were dropped during the factor analysis.

Clear Proximal Goals and Clear Navigation items merged into a single factor we named Clear Goals and Navigation. Learnability

items were cross loading with Usability of Controls and not converging into a distinct factor, so Learnability was dropped. Curiosity split into two factors, Learning and Improving Skills. Competence merged with Goal Attainability and Progress and was dropped. Progress merged into Goal Attainability and was dropped. The Elevation items (e.g. REL19/ELV01) were cross loading with Relatedness and Social Responsibility and were dropped.

Relatedness and Cooperation were merging together; specifically Cooperation was loading onto itself and cross-loading onto Relatedness. We decided to drop Relatedness and keep Cooperation because Cooperation is more specific and therefore easier to design for as a source of enjoyment. The Subversion & Lack of Real World Consequences factor and the Stakes and Investment factor both failed to converge into a single factor and were dropped.

The item ENJOY10, “Playing this game held my attention” was adapted from the Intrinsic Motivation Inventory, but conceptually overlapped too much with the definition of Focusing of Full Attention, so it was dropped to ensure content validity. ENJOY15R, “Playing this game was boring” was dropped because it was the only reverse-coded item in the EQ or SoEQ measures to survive factor analysis. Sauro and Lewis [43:204-210] discussed the impact of reverse-coding items with a negative tone and recommended avoiding including those reverse-scored items. Following this recommendation, ENJOY15R was dropped to consistently use positively worded items that did not need to be reverse-coded. ENJOY16, “I loved the feeling of what I was doing” was a bit too vague compared to the other items loading onto the Enjoyment factor and had the lowest factor loading among the remaining Enjoyment items. Removing these three items only reduced the Cronbach’s Alpha level of Enjoyment from 0.951 to 0.949.

To test the internal consistency reliability of the measures after factor analysis, the Cronbach’s Alpha was calculated for the items for each factor. Each scale had Cronbach’s Alpha levels above 0.8, indicating a high degree of internal consistency reliability. All of the items having factor loadings above [0.4] on their intended factor was evidence of convergent validity, and having no cross-loadings within 0.2 of the main loading was evidence of discriminant validity. The pattern matrix and Cronbach’s alpha levels for the measures after factor analysis can be found in Supplement E.

The psychometric properties of this version of the measures was strong, but with 246 items this measure was quite long, especially for practitioners. To create a shorter version, we chose to drop items until there were four items maximum per construct to achieve a more parsimonious and less participant-fatiguing questionnaire. This was done carefully to ensure both content validity and construct validity by ensuring that the remaining items captured the full meaning of the construct and by ensuring that a simple factor structure with high factor loadings was maintained. After items were dropped, the factor analysis was run again until a simple factor structure was reached. In the process, one of the factors, Collecting, failed to converge into a single factor and was dropped. With the conceptual similarity between Collecting and another source of enjoyment, Achievement, we considered dropping Collecting to be an acceptable to create a short version of the measures that maintained strong psychometric properties. This reduced the total number of items from 246 items to 155 items. Each scale had Alpha levels above 0.7. All but one of the scales, Loss of Self-Consciousness,

Table 1: Summary of Factors after Factor Analysis and Item Reduction.

Factors	Example Item	N of Items	Cronbach's Alpha	Lowest Factor Loading	Sources
Story	A series of events happened in the game that told a story.	4	0.957	0.815	Original
Improving Skills	My ability to play this game well improved the more that I played the game.	4	0.903	0.746	Original
Cooperation	My teammates and I worked well together as a team to achieve our objectives. The teammates could be other players in the real world or characters in the game.	4	0.930	0.799	Original
Time	I was not aware of how much time had passed until after the game was over.	4	0.927	0.825	[10, 12]; Original
Optimal Pacing	The speed of the game was just right for me – not too slow and not too fast.	4	0.916	0.740	Original
Theme	I liked the theme or topic of the game. The theme or topic of the game is the main subject matter the game focused on, such as fishing, ancient Egypt, or boxing.	4	0.899	0.671	Original
Relaxation	I had relaxing experiences.	4	0.887	0.712	[18, 44]; Original
Ability To Retry	I was able to retry what I was doing if I did not successfully complete the task.	4	0.915	0.836	Original
Humor	I burst out laughing.	4	0.917	0.715	Original
Organizing	I made things well-organized.	4	0.882	0.747	[18, 44]
Feedback	The game gave me feedback that made it really clear to me how well I was doing at playing the game.	4	0.839	0.674	[1, 32]; Original
Usability of Controls	The controls of the game were easy to learn.	4	0.891	0.592	[1, 34, 48]; Original
Focusing of Full Attention	I had my attention focused entirely on what I was doing.	4	0.909	0.736	[1, 13, 20, 32]
Autonomy	I was deciding for myself how I would take action.	4	0.863	0.606	[18, 44]
Exercise	I was moving my body enough to make my heart beat faster.	4	0.844	0.613	Original
Competition	I felt that I had more skill at playing the game than other people. The other people could be other players in the real world or characters in the game.	4	0.878	0.622	Original
Honor	What I was doing in the game was in accordance with my personal moral standards.	4	0.841	0.654	[18, 44]
Creating	I combined different parts to make new things.	4	0.911	0.702	Original
Suspense	I felt suspense because I did not know what was going to happen next.	4	0.888	0.705	[4]; Original
Vitality	I felt energized.	4	0.905	0.679	[19, 41]
Merging Of Action and Awareness	My actions in the game just seemed to happen automatically.	4	0.876	0.718	[20, 32]
Learning	I learned a lot while playing this game.	4	0.883	0.625	[18, 44]; Original
Schadenfreude-Vengeance	I got even with other players or characters in the game who offended me.	4	0.871	0.719	[6, 18, 44]; Original
Leading-Directing	I was the leader with the authority to direct the actions of others. The others could be other players in the real world or characters in the game.	4	0.911	0.740	
Sensory Pleasure and Beauty	What I saw and heard in this game was beautiful.	4	0.913	0.734	Original
Presence	I felt as if I was my character.	4	0.932	0.740	[47]
Clear Task Purpose	It was clear to me why what I was doing mattered in the game.	4	0.853	0.541	Original
Strategizing	I thought up a plan to take action in the game.	4	0.861	0.554	[8]; Original

Factors	Example Item	N of Items	Cronbach's Alpha	Lowest Factor Loading	Sources
Social Responsibility	It felt like what I was doing was benefiting society. The society could be in the game world or in the real world.	4	0.898	0.719	[18, 44]; Original
Optimal Challenge	Playing this game was challenging for me, but I was able to face that challenge	4	0.819	0.555	[1, 20, 32, 45]
Loss Of Self-Consciousness	I was not thinking about how I was presenting myself.	3	0.747	0.643	[20, 32]; Original
Achievement	I felt proud of myself when I succeeded in the game.	4	0.856	0.449	Original
Clear Goals and Navigation	It was clear to me what to do next through each step of the game.	4	0.874	0.562	Original
Sense of Control	I felt as though I had everything under control.	4	0.849	0.537	[20, 32, 44]
Goal Attainability	I felt that my current goal in the game was achievable.	4	0.888	0.610	Original
Optimal Variety	The game had a variety of new content as I went through it, but not so much variety that it was overwhelming.	4	0.823	0.440	Original
Savoring	I took the time to savor what I was experiencing.	4	0.894	0.502	Original
Task Significance	What I was doing in this game was meaningful.	4	0.887	0.431	[1]; Original
Enjoyment	I had a good time playing the game.	4	0.933	0.822	[1, 20, 28, 35]

had Cronbach's Alpha levels above 0.8, indicating a high degree of internal consistency reliability. A summary of the resulting factors can be found in Table 1 below. The full pattern matrix for the short version of the measures can be found in Supplement F.

5 LIMITATIONS AND FUTURE WORK

This study presented a new measure and preliminary validation of that measure using factor analysis. Future research may focus on using confirming the factor structure of the Enjoyment Questionnaire (EQ) and the Sources of Enjoyment Questionnaire (SoEQ) for basic and applied research in different contexts and with different populations. Future researchers wishing to do so or to replicate this study with additional samples are advised to use the longer version of the measures (Supplement E), while applied researchers may prefer the shorter version (Supplement F) to minimize participant fatigue. Researchers may also choose to use only the subscales of the SoEQ that are relevant to their research questions.

Limitations of this study include that the present study focused on content and construct validity, but the criterion validity of these measures remains to be tested. Future research may focus on testing how well these sources of enjoyment actually predict or cause increased enjoyment. Predictive studies could use regression analysis and causation studies could use controlled experiments. Because the main study presented here was conducted with only US participants, it is possible the findings in this study may not generalize to other populations or cultures. The survey focused on the last digital game participants played, so it is unknown whether or not these findings will generalize to enjoyment of non-digital games or other activities. Another limitation of this study is that the short version of the SoEQ is only one of many possible selection of items (or model specification) from the longer version of the SoEQ, so

the short version presented ought to be seen as a convenience for applied researchers and practitioners rather than the only possible short version.

6 CONCLUSION

For both basic and applied research on digital game enjoyment, researchers must keep an open mind to the possibility that their preferred theories, models, and measures may be incomplete. Having more measures in our toolbox, with factors we may not have considered before, empowers us to explore the positive experiences that make games enjoyable, and through that we may learn more about what makes life enjoyable. Hopefully the measures presented in this paper can help researchers make games, and maybe even life, a little more fun for everyone.

ACKNOWLEDGMENTS

This work was supported through a Caterpillar Fellowship Award from Bradley University. The author thanks Uday Sai Reddy Ambati and Manasa Vaidyabhushana for assistance with data analysis and visualization.

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